SET-A
Semester- I
Time- 10:00- 11:00 am
2025

St. John College of Engineering and Management

Autonomous Institute

(A Christian Religious Minority Institution)

Approved by AICTE and DTE, Affiliated to University of Mumbai / MSBTE

DTE Code : 3218 AICTE Permanent ID : 1-4790201

NAAC Accredited with Grade 'A+', Three Programs NBA Accredited

Internal Assessment Examination- II SET-B

Course Name – Engineering Physics
Course Code- 25FEBS102
Maximum Marks- 10

All Question are Compulsory
Assume Suitable data wherever required

St. John College of Engineering & Management SJCEM PALGHAR

Semester- I
Date- 13/11/25
Time-12.00 -12.30 pm

	Question	Marks	CO	Level
Q1	**Attempt all Questions**			
a	**What concept in quantum mechanics explains the non-existence of electrons within the nucleus?** a) Photoelectric effect b) Heisenberg uncertainty principle c) Quantum tunnelling d) Exclusion principle	1	3	L1
b	**Which phenomenon is responsible for the colourful patterns observed in soap bubbles?** a) Diffraction b) Polarization c) Thin film interference d) Scattering	1	4	L1
c	**What is the primary reason for the unique properties of nanomaterials compared to bulk materials?** a) Smaller particle size b) Higher density c) Larger volume d) Increased temperature stability	1	6	L2
Q2. a	What are the properties of matter waves?	3	3	L3
	or			
Q2. b	An electron is confined in a box of length 10^{-8} m. Calculate minimum uncertainty in its velocity.	3	3	L3
Q3. a	How the central dark spot is converted to bright spot in reflected system in the Newton's ring?	2	4	L2
	or			

Q3. b	Light falls normally on a soap film of thickness 5×10^{-5}cm and of refractive index 1.33. Which wavelength in the visible region will be reflected most strongly?	2	4	L2
Q4. a	Give three applications of nanomaterial.	2	6	L2
	or			
Q4. b	Define nanotechnology. What are the different types of nanomaterials?	2	6	L2

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